Why sodium cobaltate superconducts only when it is wet

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Dry Na_xCoO_2 is an ordinary metal. Slip two layers of water between its CoO_2 sheets and it superconducts near 4.5 K, and twenty years of looking for the reason inside the sheet itself have found none. We show, by density-functional calculations built on our earlier prediction of a topological electron gas at the surfaces of the layered cobaltates, that hydration performs the surface trick in the bulk. Opening the gallery between the sheets past a critical spacing near 6.4 Å does two things at once: a two-dimensional electron gas condenses between the layers, and the sodium ion's single stiff well splits into a shallow double well in which it rattles. The gas is the carrier, the rattle is the glue, and a phonon-mediated dome (peak $T_c \sim 14$ K, band 11–23 K) occupies the narrow window between the carrier-starved wall and the polaronic wall; the dry parent and the monolayer hydrate land correctly outside it. The one compound that actually superconducts holds its layers at 9.9 Å, beyond this static window; there a rigid water cage re-binds the sodium and letting that cage relax only deepens the well further, so the softening the mechanism needs must be a statistical, dynamical effect and not an adiabatic one, a picture consistent with the null hydrogen/deuterium isotope shift already on record. The evidence points at a simple rule: the pairing does not live in the CoO_2 sheets, it lives in the gallery between them.

Sodium cobaltate becomes a superconductor only when it is wet. Dry Na_xCoO_2 is an ordinary metal down to the lowest temperatures measured; hydrate it with two layers of water and it superconducts near 4.5 K [1]. That was found in 2003, and twenty years of looking for the reason inside the CoO_2 sheet itself have not produced an accepted mechanism [2].

The obvious calculation was tried early, and it failed instructively. Put static water into the lattice and relax the electrons around it, and almost nothing happens: hydration has, in the words of the sharpest of these studies, "little effect . . . outside of the predictable effects of expansion" [3], a verdict the earliest water-inclusive calculation anticipated [4] and a later one confirmed for both hydrate stages [5]. Every one of these calculations looked inside the CoO_2 sheet, and every one held the water still.

We looked somewhere else first, for a different reason. Layered cobaltates have a surface, and the same lattice that hosts a filled CoO_2 sheet also hosts a half-open gallery there. We showed, from the topology of an obstructed atomic limit seeded by an annihilating Dirac cone [6, 7], that this half-open gallery is required to carry a two-dimensional electron gas of its own, spin-split and living entirely outside the CoO_2 sheet [8]. Here we show that hydration opens the same gallery in the bulk: past a critical spacing, the charge the oxygen planes had been holding spills back into the gallery as the predicted two-dimensional gas, and at the same spacing the sodium ion, which had sat at the bottom of one stiff well, acquires a second well and begins to rattle between them. One furnishes carriers, the other a phonon-mediated bond, and a superconducting dome occupies the window between the two. The superconductivity of sodium cobaltate does not live in the CoO_2 sheets. It lives between them, which is why only water can switch it on.

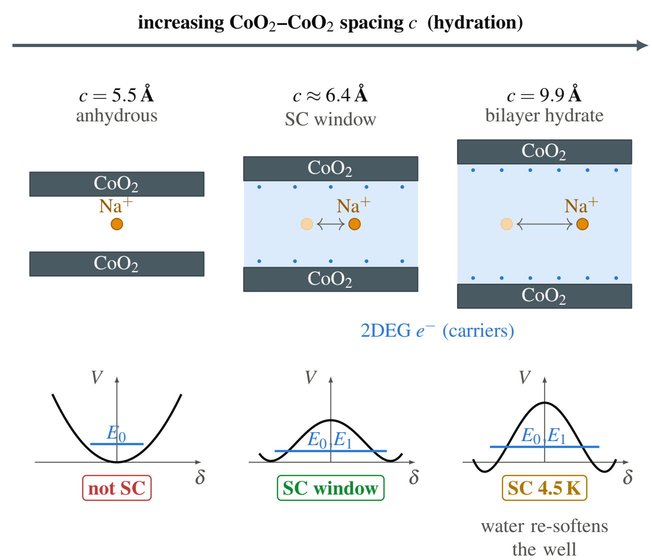

FIG. 1. The mechanism in one picture, against the gallery spacing c. Left, dry ($c = 5.5$ Å): the sodium sits in one stiff well and no gas condenses, so there is no pairing. Center, the calculated window ($c \approx 6.4$ Å): the well splits in two at the same spacing the gas turns on, and the low-lying doublet supplies the overtone boson that pairs the gas. Right, the superconducting bilayer hydrate ($c = 9.9$ Å): outside the static window, where water itself must re-soften the sodium (Fig. 4).

The transition. Picture the sodium sitting midway between two CoO_2 sheets and pull the sheets slowly apart. While the gallery is narrow, the centre is the bottom of a single well; open it far enough and the centre becomes the top of a small barrier, with one equivalent site toward each sheet, and the ion rattles between them. At the same spacing the electron count changes on its own: charge the oxygen planes had held spills into the gallery as a two-dimensional gas. We compute this directly with spin-polarized PBE and projector-augmented waves (Quantum ESPRESSO, plane-wave cutoffs 60/480 Ry), fitting $E(\delta) = E_0 + \alpha\delta^2 + \beta\delta^4$ to the total energy of the off-centre sodium at each spacing c. Full computational details, convergence tests, and the complete calculation inventory are in SM Sec. S1; the well data and fits are in SM Sec. S2. The sign of α is the whole story: positive at $c = 5.5$ Å ($\alpha = +2.5$ eV/Å^2, a single stiff well), negative by $c = 6.9$ Å ($\alpha = -0.5$ eV/Å^2, a double well), crossing at $c^*_{\mathrm{Na}} \approx 6.40$ Å. The Fermi-level density of states $N(0)$ tracks the same crossing, jumping by a factor of 30 between 5.5 and 6.9 Å and saturating beyond it, and the Bader charge returning to the sodium grows monotonically from 0 to $0.58\,e$ across the same range, an areal density of a few times 10^{13} carriers cm^{-2}, the scale of a real two-dimensional gas. These are two probes of one event: opening the gallery unbinds the electrons from the oxygen and unbinds the sodium from the centre at the same spacing, because both are the same charge leaving the same bond.

Figure 2 puts the three signatures, $\alpha(c)$, $N(0)$, and the superconducting dome built from them, on one axis with a broken scale spanning all four measured spacings. Feeding $\lambda(c)$, the boson energy, and $N(0)$ into the Allen–Dynes formula (model below, $\mu^* = 0.10$, gated on $N(0) \geq 0.1$) gives a dome pinned just above c^*, peak $T_c \approx 14$ K, that correctly excludes both anchors we can check without invoking water: the dry parent sits on the no-carrier wall and the monolayer hydrate sits past the polaron wall, and neither superconducts. Two systematics bound this number without erasing it: holding the oxygen planes fixed at one height carries a well-depth uncertainty of about $\pm 50\%$ that leaves the sign of α and every verdict in Table I untouched, and the peak height itself, 14 K over a plausible band of 11–23 K against the observed 4.5 K, states the right order of magnitude rather than a precision number. Photoemission on the parent compound finds bands roughly twice flatter than ours, which raises $N(0)$ and lowers our coupling estimate in the same proportion, so if anything the dome computed here is conservative.

The gallery band, seen directly. The band structure itself shows the same crossing. Fig. 3 tracks the Kohn–Sham spectral weight along Γ–M–K–Γ at three spacings, sodium character drawn as a blue fatband. At $c = 5.5$ Å the sodium band sits about 2 eV above E_F and carries no weight there; by $c = 9.9$ Å a band of 91% sodium character has descended through E_F, the interlayer gas predicted at the surface [6] now seen directly in the bulk gallery. The Fermi surface it leaves behind (SM Sec. S4) is a single electron barrel around Γ with no e'_g pockets, the same topology and a comparable k_F to what angle-resolved photoemission finds on the anhydrous parent [9] (composition differs, so we compare topology and size, not band identity). One further number is suggestive rather than settled: the low-energy kink photoemission sees in the parent sets in near 26 meV [9], close to the $\hbar\omega_{\mathrm{eff}} \approx 23$ meV confined sodium mode we predict once the gallery is open; the parent's gallery is closed, so this is a coincidence of scales, not our mode, and a direct test is photoemission on freshly hydrated crystals for a 20–30 meV feature on the gallery band itself.

The pairing window. We borrow the band structure and supply the coupling. The two bands sharing the gallery electron, alkali sp_z and CoO_2, are the SSH-type pair from our earlier surface model [6], split by the same Bader charge that fills the gallery band in Fig. 2(b). The sodium displacement δ is odd under reflection through the gallery midplane, so a linear electron–phonon coupling must vanish identically at the symmetric point (checked at several k_z); the leading coupling is quadratic, an electron scatters off two quanta of the rattling mode, not one, and the boson is the even overtone $\hbar\Omega = E_2 - E_0$ of the quantized well (the first excited state is parity-forbidden). The dimensionless coupling is

$$\lambda = \frac{2\,N(0)\,g^2}{\hbar\Omega}, \tag{1}$$

with $N(0)$ taken from the density-functional calculation, not fit, and g the quadratic vertex (SM Sec. S3).

The window has two walls of simple physics. Close the gallery below c^* and $N(0) \to 0$: no carriers, no pairing. Open it too far, or freeze the sodium into one well, and the ion self-traps while $\hbar\Omega$ stays near 20 meV, so λ runs past 2 into the polaronic regime and the pairing is killed there instead. Between the two walls the quantized well is soft, its effective zero-point frequency collapsing from 30 meV in the stiff single well toward a fraction of a meV in the double well (SM Sec. S6), and the Allen–Dynes dome of Fig. 2(c) sits in the resulting window, 6.35–6.45 Å, with neither the monolayer nor the bilayer anchor inside it. The last of those is the difficulty we take up now.

The 9.9 Å difficulty, and the water. The bilayer hydrate is the compound that actually superconducts, and it holds its CoO_2 layers 9.9 Å apart [10], well outside the window we just calculated. Worse, the bare sodium well at 9.9 Å is not a soft double well at all: the energy falls monotonically as the sodium slides toward one CoO_2 sheet, an instability rather than a rattler. The static picture puts the one superconducting compound in the dead polaronic zone. This is the electronic-structure literature's inert-water result restated mechanically, and the fix is to stop holding the water still.

We built the bilayer solvation cage neutron diffraction finds around the sodium [10, 11], four rigid H_2O per cell, and recomputed $E(\delta)$ against the same cell with the water deleted (construction in SM Sec. S7). Figure 4(a) is the result. The vacuum curve runs away as before; the

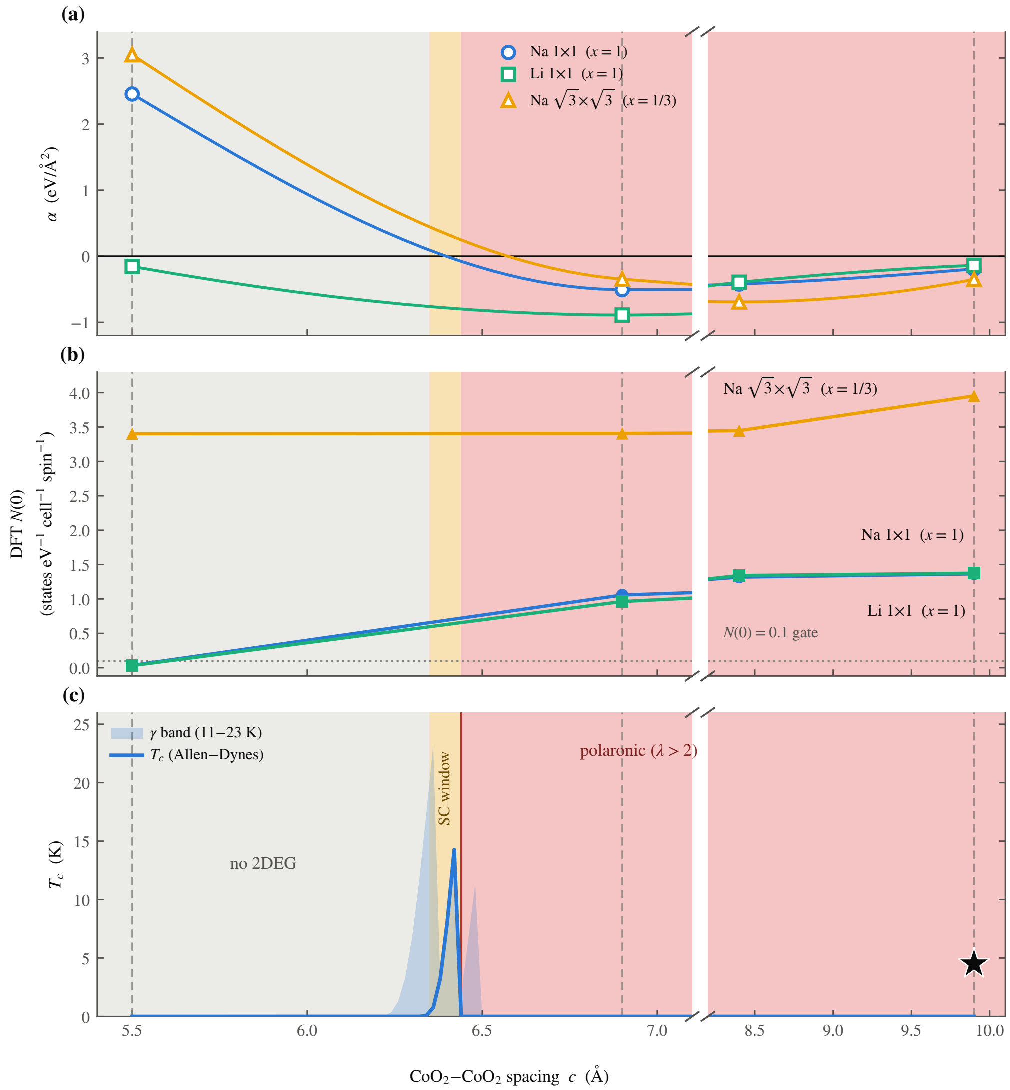

FIG. 2. The central result, against the CoO_2–CoO_2 spacing c (axis broken to place all four measured spacings on one scale). (a) The Landau coefficient $\alpha(c)$ crosses zero at $c^*_{\rm Na} \approx 6.40$ Å. (b) The Fermi-level density of states $N(0)$ switches on at the same spacing. (c) The Allen–Dynes $T_c(c)$: a narrow dome (shaded band, ansatz range 11–23 K) between the carrier wall (left) and the $\lambda = 2$ polaron wall (vertical line, right of which pairing is killed). Dashed verticals mark the anchors at 5.5, 6.9, and 9.9 Å: the dry parent and monolayer hydrate fall on the walls and are correctly not superconducting; the bilayer hydrate (star, $T_c = 4.5$ K) sits outside the static window, resolved in Fig. 4.

caged curve does not, and instead becomes a bound double well with a confined mode at $\hbar\omega_{\rm eff} \approx 23$ meV, finite and positive where the bare mode was imaginary. Water re-binds the sodium the static lattice was throwing away. Letting the same four waters relax at each sodium position does not flatten this well; it deepens it, from 199 meV frozen to 380 meV adiabatic, a genuinely stiffer mode near 37 meV, tracked by a dispersion-corrected repeat. Classical, fully relaxed water binds the sodium harder, not softer. What actually softens it is the spread among local minima the hydrogen-bond network can present at one fixed δ, at least 0.4 eV wide (Fig. 4(a), amber band), combined with a clean separation of clocks: the sodium mode has a period near 200 fs, the network reorganizes on the picosecond scale quasielastic neutron scattering measures [12]. Fast against the cage, the sodium samples the softer, instantaneous members of that ensemble rather than riding the network down to its own deep, stiff minimum, so the softening this mechanism needs is statistical, not adiabatic; the decisive open test is a finite-temperature, Born–Oppenheimer or path-integral, molecular-dynamics sampling of the coupled sodium–water system (SM Sec. S7).

This picture makes a sharp prediction about isotopes, and the prediction has already been tested. Replacing H_2O by D_2O shifts T_c by less than 0.2 K [13], and sub-

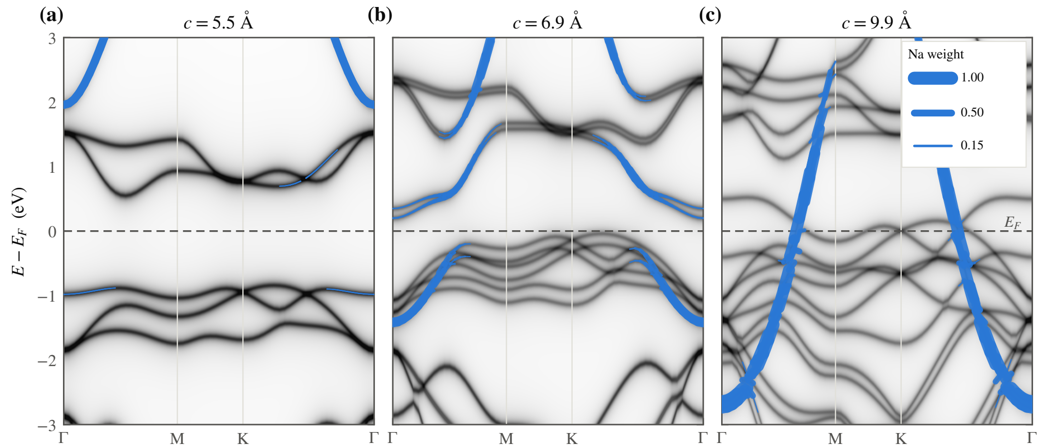

FIG. 3. The gallery band appears. Kohn–Sham spectral weight along Γ–M–K–Γ at (a) 5.5, (b) 6.9, and (c) 9.9 Å, referenced to E_F (dashed); grayscale is the total weight, the blue fatband the sodium-projected weight. At 5.5 Å the cell is a band insulator with the sodium band 2 eV above E_F; by 9.9 Å a band of 91% sodium character crosses E_F, the interlayer two-dimensional gas made visible as a single band, the payoff of the surface prediction of Ref. [6] carried into the bulk. Fermi-surface topology and the ARPES comparison are in SM Sec. S4.

stituting ^{18}O for ^{16}O is null as well [14]. A mechanism that paired electrons directly off a water vibration would not survive that result; ours requires it, because the boson is the sodium and the water's own mass barely enters the channel that gates its anharmonicity. Sodium offers no such test on itself: ^{23}Na is its only stable isotope, so nature denies a direct isotope probe of the boson, and the soft ~ 20 meV Na c-axis mode has to be sought spectroscopically instead, by inelastic neutron scattering or low-frequency Raman on fresh, hydrated crystals, where it must appear dynamical rather than as a statically split, silent ion.

Consequences. Lithium cobaltate is the same structure with a lighter ion, and it has never been made to superconduct; our picture requires that too. Its well only crosses into the double-well regime at the smallest spacing we sampled, and there it is a mere 4 meV deep while the lithium zero-point energy is 8.8 meV, so the light ion sits smeared across both minima, a quantum paraelectric with no soft anharmonic mode to offer. A heavier gallery ion should push the threshold out: for potassium the crossing moves to $c^*_{\mathrm{K}} \approx 6.75$ Å, ordering Li < Na < K with ionic size, a prediction for hydrated K_xCoO_2. Right at the carrier turn-on the gallery band is narrow enough to be Stoner-polarized, a thin magnetic strip hugging the dome (SM Sec. S8) that explains why magnetism borders the superconductivity and resolves an old contradiction, powder NMR reporting a triplet and single-crystal work a singlet, as a matter of where each sample sits relative to the Stoner boundary rather than a real disagreement.

The same charge return leaves a valence fingerprint: a site-specific probe (Co L-edge, NQR) and a charge-counting probe (titration) should disagree by $\Delta v \approx 0.06$ per cobalt only once the gallery is open, distinct from the larger valence offsets an oxonium reservoir gives [15]. Electrostatic gating of a $LiCoO_2$ film should fill the gallery to $n_{2\mathrm{D}} \approx 4 \times 10^{14}$ cm^{-2} while the light ion stays centred, a clean test of carriers without glue; Na_2CoSe_2O [16], superconducting at 6.3 K through its own Co–Se network with no occupied gallery band, is the negative control the picture asks for and gets. The pairing state we favor, a time-reversal-symmetric singlet $s_\pm$ across the gallery gas and the a_{1g} pocket, passes every constraint in the gauntlet that has eliminated most other proposals for this compound (SM Sec. S9).

What comes next. Five measurements would decide this, cheapest first: inelastic neutron scattering or low-frequency Raman for the soft, dynamical $\sim$ 20 meV sodium mode in fresh hydrate; photoemission on hydrated crystals for the gallery band and its 20–30 meV kink; XAS or NQR for the valence split; gating of $LiCoO_2$ films for carriers with the ion left centred; and hydrated K_xCoO_2 at the predicted 6.75 Å threshold. The calculation that would settle it outright is a finite-temperature, ab initio or path-integral, molecular-dynamics sampling of the coupled sodium–water system. See Supplemental Material at [URL] for the methods, the full well-transition data, the coupling model, the Fermi-surface comparison, the spin phase diagram, the water-cage construction, and the pairing gauntlet [19]. If the picture survives those tests, it states a rule wider than this compound: open a layered gallery past the spacing where its donor ion turns bistable, and the same motion that spills the ion's electrons into an interlayer gas softens the ion into the pairing boson, carriers and glue from a single structural knob.

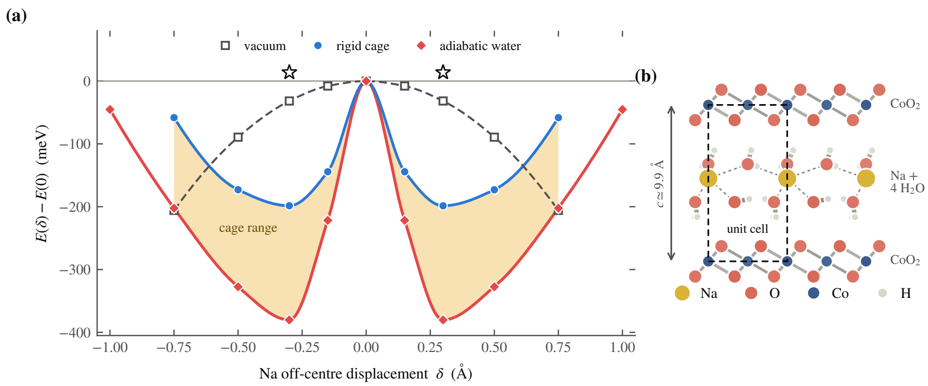

FIG. 4. The water demonstration. (a) $E(\delta)$ of the off-centre sodium at $c = 9.9$ Å: deleting the water (open squares) leaves the sodium sliding toward chemisorption; a rigid water cage (blue) rebinds it into a double well, $\hbar\omega_{\text{eff}} \approx 23$ meV; letting the cage relax (adiabatic, red) deepens the well further, to 380 meV, a stiffer 37 meV mode. The amber band is the $\gtrsim 0.4$ eV spread of cage-configurational minima at fixed δ; the open stars (one computed point, mirrored as the even curves are) mark a partially relaxed configuration caught in one shallower member of that spread. The fast sodium (~ 200 fs) samples this ensemble rather than the slow ($\sim$ps) adiabatic surface, so the softening is statistical (text; SM Sec. S7). (b) The hydrate cell along c: CoO_2 sheets, the sodium plane, and the two-layer water cage between.

TABLE I. The mechanism against the record. Verdicts: explained by the static calculation, or a prediction contingent on the dynamical water effect of Fig. 4.

Experimental fact	**Verdict**	**Source**	**Why it follows**
Anhydrous (5.5 Å) not SC	explained	[17]	$\alpha > 0$: a single stiff well (~ 30 meV) with $N(0) \approx 0$, no gallery gas.
Monolayer hydrate (6.9 Å) not SC	explained	[17]	Past the window, $\lambda \approx 20 \gg 2$: the sodium self-traps into a polaron.
Bilayer hydrate (9.9 Å) SC at 4.5 K	prediction	[1]	Static geometry is polaronic; water re-binds Na and adiabatic relaxation deepens the well further; softening into the window is statistical, the open dynamical test (SM Sec. S7).
No superconducting Li analogue	explained	—	$c^*_{\text{Li}} \lesssim 5.5$ Å and 8.8 meV zero-point energy keep the light ion centred.
Magnetic phase borders the dome	explained	[18]	The local moment turns on at the same spacing as the well transition (SM Sec. S8).
T_c is only a few K, not tens	explained	[1]	Allen–Dynes with the DFT ω_{eff} and λ peaks at 14 K (band 11–23): the right scale.
H/D substitution leaves T_c unchanged (< 0.2 K)	explained	[13]	The boson is the sodium, not the water; the water's own mass barely enters the pairing channel.

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[19] See Supplemental Material at [URL] for the DFT methods, the full well-transition data and Landau fits, the two-band coupling model and Allen–Dynes details, the Fermi-surface and ARPES comparison, the gallery band at $x = 1/3$, the mode-softening calculation, the water-cage construction and its adiabatic and dynamical analysis, the spin phase diagram and spin-orbit estimates, the pairing gauntlet, the correlation caveats, the design rule and the Li_xZrNCl null hypothesis, and the predictions beyond this compound.